Table 1 (full): game notation and parameter values

Workshop 1 report, Section 3

State, beliefs and choices	
$x^t_{k,v}$	type-k ships at node v
d	periods a ship can still stay at sea
c^t_v	control of v: $+1$ Blue, -1 Red
ℓ^t	escalation level (war at L)
w^t_r	sea state of region r
$\hat{x}^t_{-i,v}$	i's last-seen opponent count at v
$p_{i,v}$	posture: patrol (P) or assert (A)
$f^t_{k,uv}$; m^t_i	ships sailing $u \to v$; moves made
$S_{i,v}$; ϕ_v	local strength; local balance
h_v; ρ_v	incident hazard; probability
g^t_i; u_i	stage payoff; total payoff
$y_{s,w}$	MIP: ship s goes to node w
$b_{v,(n,p)}$	MIP: n hulls, posture p at v
ω; π_ω	weather scenario; its weight
V^i_v; σ_i	value of v to i; $+1$ Blue, -1 Red

Parameters (our assumptions)		
T; δ	periods (2 days each); discount	60; 0.98
N_k	fleet size	10 DDG; 8 PLAN, 6 CCG
D_k	endurance when refueled (periods)	10 DDG; 8 PLAN; 6 CCG
F_i	ship moves per period	8 Blue; 10 Red
w_k	strength per hull (CCG in rough seas)	1 DDG, PLAN; 0.6 CCG (0.3)
α	strength multiplier when asserting	1.5
κ; λ	presence damping; control rate	1; 0.3
h_0	baseline incident hazard	0.006
$h^{\mathsf{B}}_{\mathrm{A}}$; $h^{\mathsf{R}}_{\mathrm{A}}$	extra hazard if asserting	0.03; 0.036 PLAN, 0.012 CCG
L; ℓ^1	war level; starting level	5; 1
μ	chance a quiet period lowers ℓ	0.25 (D3: 0.5 at $\ell \geq 3$)
γ; η	move cost; tension cost per level	0.1; 1
W	war penalty (both sides)	200
$q_{i,v}$	detection: here / nearby / far	0.95 / 0.6 / 0.2, $\times$0.7 rough
Λ; β	MIP: incident price; persistence	20; 0.96
ε_i	escalation-risk budget	0.05 / 0.15 / 0.40
ζ_D	Blue's redesign cost per period	D1 0.3; D2 0.5; D3 0

Notation. Left: quantities that describe the game as it is played. Right: parameter values, all of them our assumptions (no public data); hazards and μ were set so that war is uncommon but not rare (0–34% of games across the configurations tested).